Plasmoids as toroidal current rings

Does Nachamkin's force-free plasmoid hold together and stop radiating? — register entry H26

kronspice simulation report · 2026-11-07 · about 8 seconds; deterministic

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Summary

J. Nachamkin's 1992 Phillips Laboratory report proposes a plasmoid made of a sphere of plasma holding a standing electromagnetic wave whose electric and magnetic fields are parallel everywhere (a force-free, or Beltrami, field), a swirling plasma flow that balances the field's pressure, and a "resonant size" at which it should stop radiating. He offers it as a model of ball lightning, with a worked example of 1 MJ in a 10 cm sphere. This example checks his field and numbers, computes the sphere's radiation lifetime, and models the two real rings his field corresponds to with kronspice: a spheromak decaying through its resistance, and an electron ring compressed by a capacitor bank.

- **The field mathematics is sound.** curl F = αF to 2×10^{-10}, div F = 0 to 10^{-11}; field lines stay on nested tori around a current ring at 0.611 of the radius (61.1 mm for R = 10 cm).
- **His density condition is ordinary plasma physics:** it is exactly the cold-plasma dispersion relation $\omega^2 = \omega_p{}^2 + \alpha^2c^2$ (to -4×10^{-17}). His Appendix B numbers are reproduced exactly.
- **His worked example breaks his own model:** the electrons would oscillate at 7.92×10^{8} m/s = 2.64 c, in a field about 1,000 times air's breakdown strength.
- **The resonant size does not stop radiation:** the plasma sphere's energy lifetime is 0.6 μs (TE) and 1.2 μs (TM), and the two halves of the force-free field need different sizes (αR = 4.4933 and 2.7437).
- **A resistive spheromak ring** of 1 MJ in 10 cm lasts 0.598 μs, 18.9 μs and 0.598 ms at 1, 10 and 100 eV; lasting one second needs 14082 eV (163 million K).
- **A collapsing electron ring** driven by a 50 kJ bank goes from 16.4 cm to 2.60 cm and 2.0 to 15.0 MeV, as R proportional to $B^{-½}$ predicts; it takes 21 J of the bank's energy and the ledger closes to 10^{-9}.

What is being tested

The claim (hypothesis H26)

Long-lived plasmoids (ball lightning) are force-free, time-harmonic electromagnetic fields in a plasma, stabilized by vortical flow and non-radiating at a resonant size (Nachamkin, *Force-Free Time-Harmonic Plasmoids*, PL-TR-92-3044, 1992).

The standard prediction beside it

A standing wave in a cold plasma obeys $\omega^2 = \omega_p{}^2 + \alpha^2c^2$, and a finite plasma sphere with no outside feed has only leaky (radiating) modes. At zero frequency the same l = 1 field is the spheromak, a laboratory force-free current ring that decays through its resistance (μs to ms in the lab). A ring of electrons in a rising magnetic field is the electron-ring compressor of the electron-ring accelerators, whose radius follows R proportional to $B^{-½}$ from conservation of canonical angular momentum. The example sits beside the other electron-ring models: *kron_electron* (H21), *space_vortex* (H13) and *photon_electron* (H23).

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Results

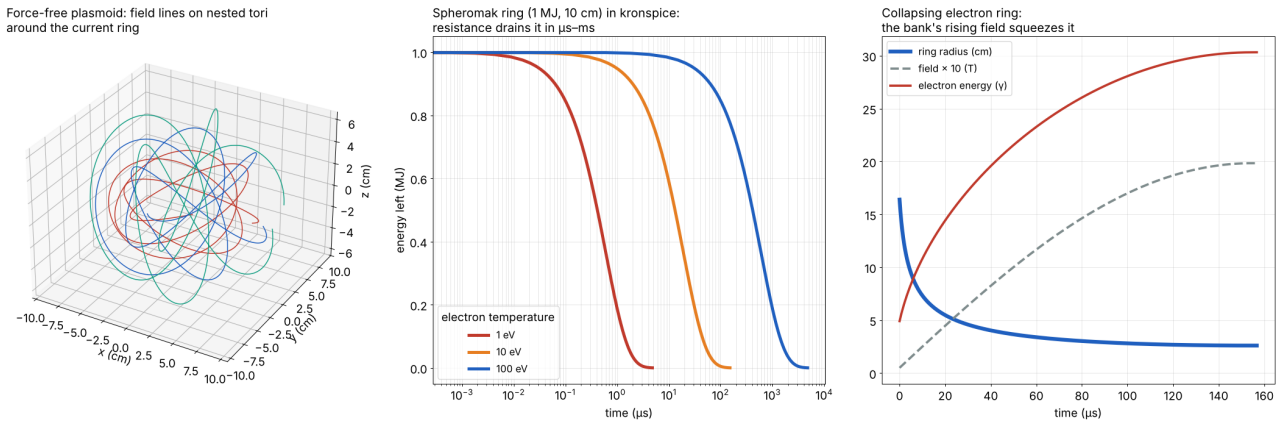


Figure 1. Left: field lines of Nachamkin's l = 1 force-free field, lying on nested tori around the current ring (section 1). Middle: the 1 MJ, 10 cm spheromak ring in kronspice, energy left against time on a log axis at electron temperatures 1, 10 and 100 eV (section 4). Right: the collapsing electron ring, ring radius (cm), field × 10 (T) and electron energy (γ) against time as the bank's field rises (section 5).

1. The force-free field

His l = 1 field (eq. 8) is identical to the spheromak field. It obeys curl F = αF to 2×10^{-10} and div F = 0 to 10^{-11}. A traced field line keeps its poloidal flux ψ to 10^{-14}, so it stays on its torus, and the tori nest around a current ring at 0.611 of the sphere's radius (61.1 mm for R = 10 cm).

2. His critical frequency is the cold-plasma cutoff

Appendix B recomputed (R = 10 cm, wavenumber k = 2686 /m):

Quantity	Here	Nachamkin
Free charge density $\rho = (m/e)\varepsilon_0 c^2(k^2 - \alpha^2)$	−32.63 C/m³	−32.63 C/m³
Free electrons	8.532×10^{17}	8.532×10^{17}
Field E_0 = C·R·ω²/|e/m| for his C = 0.01 at 126 GHz	3.566×10^{9} V/m	3.566×10^{9} V/m

- That density's plasma frequency ω_p satisfies $\omega^2 = \omega_p^2 + \alpha^2c^2$ to -4×10^{-17}: the plasmoid is a standing wave of a cold plasma with wavenumber α, and his critical frequency $\omega_0 = \alpha c$ is that wave's cutoff.
- At his chosen field the electrons oscillate at ω·C·R = 7.92×10^{8} m/s = 2.64 c, above the speed of light, so his non-relativistic electron model does not hold for his own example. The field is also about 1,000 times air's breakdown strength.

3. Does the resonant size stop radiation?

Exact l = 1 resonances of a plasma sphere with his radius and density, with outgoing waves only (nothing feeding it from outside):

Part of his field	Frequency	Size it needs (αR)	Q	Energy lifetime
TE (his M part)	128.1583 GHz	4.4933 (his choice)	4.8×10^{5}	6×10^{-7} s
TM (his N part)	128.1471 GHz	2.7437	9.4×10^{5}	1.2×10^{-6} s

- The plasmoid radiates: its energy leaks away in about a microsecond. At 1 MJ the initial radiated power is about 2×10^{12} W.
- The two halves need different sizes. An equal mix of them, which is what makes E parallel to H, is not one free mode of the sphere; holding it needs outside driving or surface currents, the weak point Nachamkin names himself.

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4. The spheromak ring in kronspice

At zero frequency the field is a force-free current ring: there is no Lorentz force on the current, so the ring neither pinches nor bursts, and electrical resistance ends it. The field decays as one mode with lifetime $\mu_0/(\eta\alpha^2)$, with η the Spitzer resistivity. As a circuit the ring is an inductor $L = 2W/I^2$ carrying the ring current I, in series with a resistor $R = P/I^2$ (W the field energy, P the ohmic power). For 1 MJ in a 10 cm sphere: field scale $B_0 = 80$ T, ring current 7.43 MA, magnetic pressure 2.5×10^4 atm, held only by a wall or an outside field.

Electron temperature	Ring L	Ring R	Energy lifetime (theory)	kronspice	Heat
1 eV	3.62×10^{-8} H	0.0303 Ω	5.98×10^{-7} s	5.98×10^{-7} s	9.997×10^5 J of 10^6 J
10 eV	3.62×10^{-8} H	0.000957 Ω	1.89×10^{-5} s	1.89×10^{-5} s	9.997×10^5 J of 10^6 J
100 eV	3.62×10^{-8} H	3.03×10^{-5} Ω	0.000598 s	0.000598 s	9.997×10^5 J of 10^6 J

To last one second the ring would need T = 14082 eV (163 million K), fusion-plasma temperatures. Ball lightning reported to last seconds cannot be a resistive spheromak in air.

5. A collapsing electron ring, driven by kronspice

Declared setup: a 1000 μF bank at 10 kV is discharged into a 10 μH solenoid with 16 turns per metre (kronspice solves this circuit); 2 MeV electrons circle the axis in a 0.05 T bias field. As the coil field rises, the induced electric field speeds the electrons up and the stronger field bends them tighter, so the ring collapses.

Quantity	Start	End	Law
Field	0.050 T	1.984 T (in a quarter period)	
Ring radius	16.4 cm	2.60 cm	R proportional to $B^{-½}$: 2.60 cm
Momentum	1	×6.3007	×6.2989
Electron energy	2.0 MeV	15.0 MeV	
Ring current (10^{13} electrons)	456 A	2933 A	

- The electrons' angle round the axis is a cyclic coordinate (as in *kaluza_kron*), so their canonical angular momentum is conserved, here to 6×10^{-4}.
- The work of the induced field, 2.0808×10^{-12} J per electron, equals the energy gain, 2.0808×10^{-12} J.
- The bank (5×10^4 J) drives the coil and the ring takes 21 J of it. The "void" inside the ring is field, not a source: every joule the ring gains comes from the bank.

Ledger and validation

- Spheromak decay: the kronspice energy lifetime equals the theoretical $\mu_0/(\eta\alpha^2)$ at all three temperatures, and the heat (9.997×10^5 J) accounts for the 10^6 J stored.
- Electron ring: kronspice ledger -10^{-9}; induced-field work equals the electron energy gain.
- Program conclusion: no energy appears from the plasmoid or the void inside the ring; radiation, resistance and the bank account for all of it.
- *main_test.go* checks the field, Appendix B, the modes, the spheromak decay and the ring compression.

Limits of the model

- The plasma sphere's resonances are computed for a uniform cold plasma with only outgoing waves; outside driving, surface currents and the swirling flow are not modelled.
- The spheromak is treated as one resistive decay mode with Spitzer resistivity; the wall or outside field that would hold its 2.5×10^4 atm pressure is not modelled.
- Nachamkin's own suggested extra ingredients (quantum effects, outside fields) have no equations in the paper and are not modelled.

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Status

Register entry H26. The paper's field mathematics is sound and its electron density follows from standard plasma physics. Two parts fail: the worked example needs electron quiver speeds of 2.6 c, so the non-relativistic model breaks; and the resonant size does not stop radiation (energy lifetime about 1 μs), with the TE and TM halves needing different sizes. A 1 MJ, 10 cm spheromak lives 19 μs at 10 eV and needs 14 keV to last 1 s — not ball lightning in air. Long-lived ball-lightning plasmoids, if electromagnetic, need something the paper lacks. The real, measured force-free rings are spheromaks and compressed electron rings, both modelled here with kronspice.

Reproducing the results

```
go run ./examples/plasmoid_rings -viz examples/plasmoid_rings/viz   # about 8 seconds
python3 examples/plasmoid_rings/viz/render.py
go test ./examples/plasmoid_rings
```

Sources: J. Nachamkin, PL-TR-92-3044 (1992), DTIC AD-A257 765; S. Chandrasekhar, P. C. Kendall, *Astrophys. J.* 126, 457 (1957); M. N. Rosenbluth, M. N. Bussac, *Nucl. Fusion* 19, 489 (1979); F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion*; J. A. Stratton, *Electromagnetic Theory* (1941), §9.22; NRL Plasma Formulary; V. I. Veksler, Proc. CERN Symposium on High Energy Accelerators (1956).

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